

Yuehong Cassandra Tai

✉ yhcasstai@psu.edu 🌐 cassyyuehtai.netlify.app

ACADEMIC APPOINTMENTS

Pennsylvania State University

Assistant Research Professor, Center for Social Data Analytics 2025-

Assistant Director, Center for Social Data Analytics

Postdoctoral Scholar, Center for Social Data Analytics 2022-2024

EDUCATION

Ph.D. Program in Political Science, University of Iowa, U.S.A. 2017-2022

Fields: Comparative Politics, Political Methodology, American Politics

MSc in International Political Economy, Nanyang Technological University, Singapore 2008

LL.B. in Diplomacy, Renmin University of China, China 2007

RESEARCH AREAS

Computational Social Science - Generative AI Evaluation - Measurement and Validity - Political Communication - AI and Democratic Accountability - Survey, Text, Speech, and Image as Data.

PUBLICATIONS

AI and Computational Social Science

18. **Tai, Yuehong Cassandra**, Yu-Ru Lin, and Bruce A. Desmarais. 2026. “Public Officials’ Online Sharing of Misinformation: Institutional and Ideological Checks.” *Political Communication*.
17. Yu, Dong Erico, and **Yuehong Cassandra Tai**. 2026. “Speak ‘As Good As Men’? An Empirical Study of Gender Variation in Chinese Courtroom Audio” *Accepted at Political Studies Review*.
16. **Tai, Yuehong Cassandra**, Nitheesha Nakka, Khushi Navin Patni, Sarah Rajtmajer, Kevin Munger, Yu-Ru Lin, and Bruce A. Desmarais. 2025. “The Digitally Accountable Public Representation Database: Measuring Online Communication by Federal, State, and Local Officials.” *Nature Scientific Data*.
15. **Tai, Yuehong Cassandra**, Khushi Navin Patni, Nicholas Daniel Hemauer, Bruce A. Desmarais, and Yu-Ru Lin. 2025. “GenAI vs. Human Fact-Checkers: Accurate Ratings, Flawed Rationales.” In *Proceedings of the 17th ACM Web Science Conference 2025*.
14. Nakka, Nitheesha, Issac Pollert, Lingyu Fuca, and **Yuehong Cassandra Tai**. 2025. “Navigating Hate and Anti-Hate Speech: Bridging Large Language Model and Human Expertise in Public Officials’ Online Communication.” [with graduate students] *Workshop Proceedings of the 19th International AAAI Conference on Web and Social Media*.
13. Biswas, Ahana, Yu-Ru Lin, **Yuehong Cassandra Tai** and Bruce A. Desmarais. 2025. “Political Elites in the Attention Economy: Visibility Over Civility and Credibility?” In *Proceedings of the International AAAI Conference on Web and Social Media*.

12. **Tai, Yuehong Cassandra**, Roan Buma and Bruce A. Desmarais. 2023. “Official Yet Questionable: Examining Misinformation in U.S. State Legislators’ Tweets.” *Journal of Information Technology & Politics*.

Public Opinion and Measurement

11. **Tai, Yuehong Cassandra** and Frederick Solt. 2026. “Trust in Civil Servants: A Cross-National Dataset for Public Policy Research, 1986–2022” *Accepted at Political Studies*.
10. Hu, Yue, **Yuehong Cassandra Tai**, and Frederick Solt. 2026. “Data Wrangling Before Harmonization: Best Practices for Getting Past the ‘Janitor Work.’” *Accepted at PS: Political Science & Politics*.
9. Ma, Haofeng, Jeongho Choi, **Yuehong Cassandra Tai**, Yue Hu, and Frederick Solt. 2026. “Macrodiscontent Across Countries” *Accepted at European Political Science Review*
8. **Tai, Yuehong Cassandra**. 2026. “Introducing the Trust in Government (TrustGov) Dataset: A New Resource for Cross-National Time-Series Trust Research” *Research & Politics*.
7. **Tai, Yuehong Cassandra**. 2025. “The Upsides and Downsides of Generalized Trust: The Mediating Effect of Democratic Institutions.” *International Journal of Public Opinion Research*.
6. Hu, Yue, **Yuehong Cassandra Tai**, Hyein Ko, Byung-Deuk Woo, and Frederick Solt. 2025. “An Incomplete Recipe: One-Dimensional Latent Variables Do Not Capture the Full Flavor of Democratic Support.” *Research & Politics*.
5. Woo, Byung-Deuk, Hyein Ko, **Yuehong Cassandra Tai**, Yue Hu, and Frederick Solt. 2024. “Public Support for Gay Rights Across Countries and Over Time.” *Social Science Quarterly*.
4. Hu, Yue, **Yuehong Cassandra Tai***, and Frederick Solt. 2024. “Revisiting the Evidence on Thermostatic Response to Democratic Change: Degrees of Democratic Support or Researcher Degrees of Freedom?” *Political Science Research and Methods*. [* co-first author]
3. **Tai, Yuehong Cassandra**, Yue Hu, and Frederick Solt. 2022. “Democracy, Public Support, and Measurement Uncertainty” *American Political Science Review*.

Chinese Politics

2. **Tai, Yuehong Cassandra**, Jielu Yao and Elise Pizzi. 2022. “Policy Adoption and Diffusion during the COVID-19 Crisis” *Journal of Asian Public Policy*.
1. **Tai, Yuehong Cassandra**. 2022. “Determinants of Chinese Provinces’ Responses to the 2018 Vaccine Scandal: Policy Orientation and Neighboring Effect” *Asian Survey*.

GRANTS

Center for Socially Responsible Artificial Intelligence, “Transparent and Reproducible AI-assisted Annotation for Social Data: A Multi-Agent framework,” effective Dec 20, 2025 and expires June 30, 2027. Co-PI, Qunhua Li. \$25,000.

WORKING PAPERS

2. Ko, Hyein, **Yuehong Cassandra Tai**, and Nora Webb Williams “Can We Trust LLM-Generated Data? The CRAFT Framework for Measurement and Inference in Political Science” [Co-lead Author] [Under Review]
1. **Tai, Yuehong Cassandra**, Xun Cao, Yu-Ru Lin, and Bruce A. Desmarais. “Where Legislators Speak Changes What They Say: Platform Conditions Climate Attention and Framing” [Under Review]

SELECT WORKS IN PROGRESS

8. “When F1 Lies: Taking Constructs Seriously in AI-Assisted Measurement” (Qunhua Li)
7. “Who Speaks for AI? Local News Papers and AI Policy News Frames” (with Hyein Ko)
6. “In Name Only: What LLM Annotators Learn from Codebooks, Examples, and Fine-Tuning?” (with Viraj Mishra, Shubham Gupta, Dhruv Gupta, and Qunhua Li) [**with undergraduate students**]
5. “Large Language Models in Social Science: A Literature Review of LLMs and GenAI in Political Science and Communication” (with Pooja Iyer)
4. “Deliberation About AI in Local Government”
3. “AI and democratic legitimacy” (with Jia Li, M. Rosemary Pang, Erico Yu)
2. “How Does Public Opinion Affect Climate Change Policies?” (with Xun Cao and Fred Solt) [**Lead Author**]
1. “Surfing Waves or Making Waves? Declining Trust in Democracy and Election of Personalist Leaders” (with Jia Li and M. Rosemary Pang)

CONFERENCE PRESENTATIONS

- “When F1 Lies: Taking Constructs Seriously in AI-Assisted Measurement”
PolMeth 2026, Michigan State University July 2026
- “Can We Trust LLM-Generated Data? The CRAFT Framework for Measurement and Inference in Political Science”
AI and Social Research: Empathic AI, Metascience, and Methodology, Penn State June 2026
- “When F1 Lies: Taking Constructs Seriously in AI-Assisted Measurement”
AI for Social Science Research Methods, Yale May 2026
- “Who Speaks for AI? Local News Papers and AI Policy News Frames”
Midwest Political Science Association Conference April 2026
- “From Annotation to Oversight: A Pipeline Approach to GenAI Evaluation”
Southern Political Science Association Conference Jan 2026
- “Who Speaks, Who Falls Silent: Strategic Climate Messaging by State Legislators on Facebook and X”
Protecting the Election: AI & Governance, Washington University in St.Louis, MO Oct 2025
- “Who Speaks, Who Falls Silent: Strategic Climate Messaging by State Legislators on Facebook and X”
PolNet-PaCSS 2025, Harvard University, MA Aug 2025

PolMeth 2025, Emory University, GA	July 2025
“State Legislators’ Misinformation and the Limits of LLMs’ Evaluative Reasoning”	
Information and Misinformation in Elections, California Institute of Technology, CA	Jan 2025
“The Digitally Accountable Public Representation Database: Measuring Online Communication by Federal, State, and Local Officials”	
(Large) Language Models in Social Science 2024, University of Pennsylvania, PA	Sep 2024
PolMeth 2024, University of California, CA	July 2024
“Climate Clash: Fossil Fuel Influence, Public Opinion, and State Legislators’ Online Discourse”	
APSA 2024, University of Pennsylvania, PA	Sep 2024
Visions in Methodology 2024, Washington University in St.Louis, MO	May 2024
“Understanding Asian Hate in State Legislatures: An Electoral Connection”	
State Politics and Policy Conference 2024, University of Virginia, VA	June 2024
“The Prevalence of Misinformation in U.S. State Legislators’ Facebook Posts”	
American Political Science Association Conference 2023, LA	Sep 2023
“Dynamic Trust and States’ Failure in Public Health: Evidence from Global Time Series Data”	
American Political Science Association Conference 2023, LA	Aug 2023
“The Prevalence of Misinformation in U.S. State Legislators’ Facebook Posts”	
State Politics and Policy Conference 2023, University of Pittsburgh, PA	May 2023
“How Does Public Opinion Affect Climate Change Policies? Constructing Measures of Climate Change Public Concern and Testing Their Effects on Policy Outputs”	
Political Economy and International Relation Workshop, PSU	Mar 2023
“The Upsides and Downsides of Generalized Trust: The Mediating Effect of Democratic Institutions and Trust in Implementing Institutions”	
Southern Political Science Association Conference	Jan 2023
“Democracy, Public Support, and Measurement Uncertainty”	
C-SoDA Brownbag Series	Oct 2022
“Examining States’ Public Health Failures by Measuring Trust Cross Countries and Over Time”	
C-SoDA Kickoff Poster Presentation	Sep 2022
“Dynamic Trust and States’ Failure in Public Health: Evidence from a Global Time Series Data”	
Midwest Political Science Association Conference	Apr 2022
Iowa Women Studying Politics (IWSP) Conference	Feb 2022
“Policy Burden, Issue Salience, and COVID-19 Policy Adoption in China”	
APSA Chinese Politics Mini-Conference	Oct 2021
“Democracy, Public Support, and Measurement Uncertainty”	
Joint Quantitative Political Science Conference for Asia and Australasia	Jan 2021
“Policy Orientation and Diffusion in Responsive Governance: Responses to the Chinese 2018 Vaccine Scandal.”	
Association of Chinese Political Studies Annual Meeting	Nov 2020
“Who is Influencing Public Opinion during a Public Health Crisis on Social Media in China?”	

TEACHING EXPERIENCE**Invited Talk and Guest Lecture****Guest Speaker, Penn State University**

Social Science Research Institute Open House

Fall 2025

Hands-On with AI: Evaluating and Applying Models for Social Science Data

Guest Speaker, Penn State University

Social Science Research Institute Open House (Open to Faculty Members in SSRI) Spring 2025

How to Navigate High Performance Computing

Guest Lecturer, Penn State University

Approaches and Issues in Big Social Data (Graduate Level)

Spring 2025

GenAI vs. Human Fact-Checkers: Accurate Ratings, Flawed Rationales

Guest Lecturer, University of Massachusetts Amherst

Data Analytics and Computational Social Science (Graduate Level)

Fall 2024

LLMs in social science

Guest Speaker, Xi'an Jiaotong University, China

Summer Speaker Series (Graduate Level)

Summer 2023

Latent Variable Measurements

Guest Lecturer, Penn State University

Approaches and Issues in Big Social Data (Graduate Level)

Spring 2023

Measurement

Guest Lecturer, University of Iowa

Advanced Methods Seminar (Graduate Level)

Fall 2021

Machine Learning

Analyzing Political Data (Undergraduate Level)

Spring 2021

P-Value & Multiple Regression

Public Opinion

Public Opinion and Democracy

Fall 2021

National Security Policy

Cyber Security

March 2020

Workshop, East China Normal University, Shanghai

Introduction to Python (Graduate Level)

Dec 2020

Web scraping, social media data wrangling and analysis

Method Courses**Instructor, Penn State University**

C-SoDA Research Practicum: AI and Multimodal data (Undergraduate Level)

Spring 2026

Instructor, Penn State University

C-SoDA Research Practicum: AI and Governance (Undergraduate Level)

Fall 2025

Instructor, Penn State University

PLSC 10: Scientific Study of Politics (Undergraduate Level) Fall 2024

Instructor, Penn State University

PLSC 308: Introduction to Political Research (Undergraduate Level) Spring 2024

Instructor, Penn State University

SoDA 308: Research Design for Social Data Analytics (Undergraduate Level) Fall 2023

Independent Instructor, University of Iowa

Hawkeye Poll (Undergraduate Level) Spring 2022

Guided students to design survey questions on both national and local politics, carry out telephone interviews, analyze and interpret data, and write release press and analytical papers.

Teaching Assistant, University of Michigan

Bayesian Modeling for the Social Sciences II: Advanced Topics, ICPSR Program, Summer 2021

Graded students' assignments with detailed comments and offered one-on-one consultations on students' individual projects.

Comparative Politics and American Politics Courses

Teaching Assistant, University of Iowa

Introduction to Political Behavior Fall 2019

Introduction to Comparative Politics Fall 2018, Spring 2019

The Presidency Fall 2017

Introduction to Comparative Politics (Online Course) Summer 2018, 2019

EXTERNAL AND CAMPUS SERVICE

Member: American Political Science Association (sections on Political Methodology), Midwest Political Science Association, and Southern Political Science Association

Reviewer: American Political Science Review, American Journal of Political Science, British Journal of Political Science, Research and Politics, The International Conference on Weblogs and Social Media (ICWSM), International World Wide Web Conference 2025 (WWW2025), International World Wide Web Conference 2026 (WWW2026), Humanities and Social Sciences Communications, Public Opinion Quarterly, Asian Survey, International Journal of Public Opinion Research, International Journal of Comparative Sociology.

Seed Grant Reviewer ("Analytical and Social AI" program): Social Science Research Institute. Pennsylvania State University 2026

AI Panel Facilitator: 2026 Data Summit. Pennsylvania State University 2026

Seed Grant Reviewer: Center for Social Data Analytics. Pennsylvania State University 2022, 2023, 2024, 2025, 2026

POLMETH Mailing List Editor: The Society for Political Methodology. 2024 -

Planning Committee Member: 16th Postdoctoral Research Symposium. Pennsylvania State University 2023

Committee Member: Postdoctoral Scholar Committee. College of the Liberal Arts, Pennsylvania State University 2022

iPeer Mentor: iPeer Mentorship Program. International Student and Scholar Services, University

of Iowa 2021

Discussant: Government Transparency in China. Midwest Political Science Association Annual Meeting Apr 2021

Session Chair and Discussant: Time and Space. Political Science Methodology Conference for Asia and Australasia Jan 2021

Co-Host: *2020 Iowa Caucus Conference*. University of Iowa Feb 2020

Session Chair: Understanding Polarization, COVID & Computation and Social Media & Politics. Politics and Computational Social Science and Political Networks Aug 2020

Discussant: Chinese Politics. Southern Political Science Association Conference Jan 2020

AWARDS & HONORS

Nominated to Global Student Awards, University of Iowa Fall 2021

Finalist for annual “Thesis Challenge,” University of Iowa Fall 2021

Graduate Post-Comp Research Fellowship (\$10,500), University of Iowa Fall 2021

Graduate College Summer Fellowship (\$5,000), University of Iowa Summer 2021

Department Research Grant (\$1,000), University of Iowa Spring 2020

GSS Travel Funds (\$1,000), University of Iowa Spring 2020

Department Research Grant (\$300), University of Iowa Spring 2019

Center for Asian and Pacific Studies Graduate Travel Grant (\$500), University of Iowa Fall 2019

Department Summer Training Grant (\$2400), University of Iowa Summer 2018

ADDITIONAL TRAINING AND SKILLS

Courses Taken in Graduate School

Methodology and Quantitative Analysis: Introductory, Intermediate, and Advanced Methodology; Modeling Time and Space; Factor Analysis and Structural Equation Models

Computational and Data Science Methods: Computational Methods for Comparative Politics; Text as Data; Introduction to Informatics; Network Analysis; Bayesian Analysis, Database Management

Training

Summer Institutes in Computational Social Science, Rutgers, NJ 2024

Certificate

Certificate in Informatics, The University of Iowa, U.S.A. 2021

Programming in Python, Java, and SQL.

ICPSR, University of Michigan, Ann Arbor, MI 2018

Causal Inference for the Social Sciences & Multilevel Models II: Advanced Topics.

Skills

Analysis & Programming: R, Python, SQL, Stata, Stan, JAGS, Mplus, Java.

Application: LaTeX, Markdown, Git(GitHub).

Language: English, Chinese (Mandarin)

REFERENCES

Frederick Solt, Ph.D

Bruce A. Desmarais, Ph.D

frederick-solt@uiowa.edu; (319)335-2340

Professor, University of Iowa

Xun Cao, Ph.D

xuc11@psu.edu; (814) 865-8749

Professor, Penn State University

Elizabeth J. Menninga, Ph.D

elizabeth-menninga@uiowa.edu; (319) 335-3835

Associate Professor, University of Iowa

bdesmarais@psu.edu; (814) 863-1595,

Professor, Penn State University

Bryce Dietrich, Ph.D

bjdietri@purdue.edu; (765) 494-4161

Associate Professor, Purdue University